MACHINE LEARNING SUPERVISED & UNSUPERVISED LEARNING

HISTORY OF MACHINE LEARNING

1950

Turing Test

Alan Turing proposes the "Imitation Game" — the first framework for machine intelligence.

1956

AI is Born

Dartmouth Conference coins "Artificial Intelligence." McCarthy, Minsky, Shannon, Simon attend.

1957

Perceptron

Frank Rosenblatt creates the Perceptron — first trainable neural network for classification.

1967

Nearest Neighbor

K-Nearest Neighbor algorithm developed — the first pattern recognition algorithm.

1986

Backpropagation

Rumelhart & Hinton popularize backprop — enabling multi-layer neural network training.

HISTORY OF MACHINE LEARNING

1995

SVM Invented

Support Vector Machines introduced by Vapnik — powerful for classification with small data.

2001

Random Forest

Breiman introduces Random Forests — ensemble method combining many decision trees.

2006

Deep Learning

Hinton coins "Deep Learning." Breakthrough in training deep neural networks with GPUs.

2012

ImageNet Win

AlexNet wins ImageNet by a huge margin — marks beginning of the deep learning era.

2017+

Transformers & LLMs

Attention mechanism leads to GPT, BERT, ChatGPT — modern large language models.

Recent Milestones:

- AlphaGo beats world champion
- Self-driving cars (Tesla, Waymo)
- ChatGPT reaches 100M users in 60 days
- AI generates art, music, code & video

WHAT IS MACHINE LEARNING?

> *"Machine Learning is a field of AI that gives computers the ability to learn from data and improve their performance on tasks without being explicitly programmed."*
>
> — Arthur Samuel, 1959

⚙️ Traditional Programming

- ✗ You write explicit rules (if-then-else)
- ✗ Rules must cover EVERY possible case
- ✗ Input + Rules → Output
- ✗ Hard to scale; breaks with exceptions
- ✗ Example: if score > 90: grade = "A"

🤖 Machine Learning

- ✓ Algorithm learns rules FROM data
- ✓ Improves automatically with more data
- ✓ Input + Output → Algorithm learns rules
- ✓ Scales well; handles complex patterns
- ✓ Example: model trains on 10,000 grade records

THREE TYPES OF MACHINE LEARNING

Supervised Learning

- Uses labeled training data
- Learns input → output mapping
- Goal: predict for new inputs
- Examples: spam filter, loan approval
- Subtypes: Classification, Regression

Unsupervised Learning

- No labels — finds hidden patterns
- Discovers structure in data
- Goal: understand groupings
- Examples: customer segments, topic modeling
- Subtypes: Clustering, Dim. Reduction

Reinforcement Learning

- Agent learns through trial & error
- Receives rewards / penalties
- Goal: maximize cumulative reward
- Examples: game AI, robotics, trading
- Key concept: policy, environment, reward

★ Today's Focus: Supervised & Unsupervised Learning

SUPERVISED LEARNING

LEARNING WITH LABELS

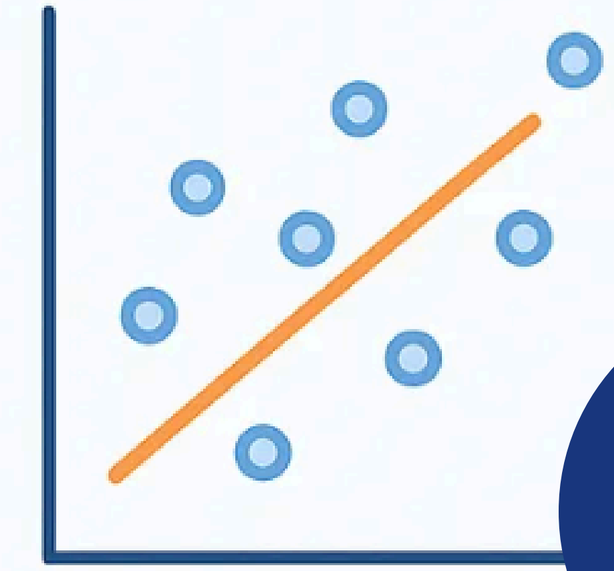
The algorithm learns a mapping function f(x) = y, where x is the input and y is the known output label, then generalizes to predict y for unseen x values.

📂 Classification

- Output is a discrete category/class
- Binary: Yes/No, Spam/Not Spam, 0/1
- Multi-class: Cat / Dog / Bird / Fish
- Algorithms: Logistic Regression, Decision Trees, SVM, KNN, Naive Bayes, Neural Nets
- Metrics: Accuracy, Precision, Recall, F1
- Use case: Email spam, medical diagnosis, image recognition, credit risk

📈 Regression

- Output is a continuous numeric value
- Predicts quantities, amounts, measurements
- Example: Predict house price, temperature
- Algorithms: Linear Reg, Polynomial Reg, Ridge, Lasso, Random Forest Regressor, SVR
- Metrics: MSE, RMSE, MAE, R^2 Score
- Use case: Stock prices, salary prediction, demand forecasting, weather



Regression

Continuous prediction

Classification

Distinct class labels

SUPERVISED LEARNING WORKFLOW

STEP 1

Collect Labeled Data

Gather data with known correct answers (labels). Quality > quantity.

STEP 2

Preprocess & Clean

Handle missing values, normalize features, encode categories.

STEP 3

Split Dataset

Train set (70-80%) for learning; Test set (20-30%) for evaluation.

STEP 4

Train Model

Algorithm learns patterns from training data by minimizing loss.

STEP 5

Evaluate & Tune

Measure accuracy, precision, recall on test data. Tune hyperparams.

STEP 6

Deploy & Monitor

Serve model in production. Monitor drift and retrain as needed.

SUPERVISED LEARNING

KEY SUPERVISED LEARNING ALGORITHMS

Linear Regression

Regression

Fits a straight line y = mx + b to data. Best for linear relationships between variables.

e.g. House price prediction by size

Logistic Regression

Classification

Uses sigmoid function to output probability (0–1). Despite name, it is a classifier.

e.g. Disease diagnosis (positive/negative)

Decision Tree

Both

Splits data using if-then rules at each node. Highly interpretable but prone to overfitting.

e.g. Loan approval: income > 50k AND age > 25?

K-Nearest Neighbors

Both

Classifies a new point by the majority vote of its K closest neighbors in feature space.

e.g. Movie recommendation by similar users

Support Vector Machine

Both

Finds the optimal hyperplane maximizing the margin between classes. Uses kernel trick for nonlinear data.

e.g. Face recognition, text classification

Random Forest

Both

Ensemble of many decision trees. Averages results to reduce overfitting. Robust and accurate.

e.g. Credit fraud detection

Naive Bayes

Classification

Probabilistic classifier based on Bayes theorem. Assumes features are independent. Very fast.

e.g. Spam filter, sentiment analysis

Neural Networks

Both

Layers of connected nodes inspired by the brain. Learns complex patterns. Basis of deep learning.

e.g. Image recognition, speech-to-text

Gradient Boosting

Both

Builds trees sequentially; each tree corrects errors of the previous. XGBoost is a popular variant.

e.g. Kaggle competition winner; tabular data

OVERFITTING, UNDERFITTING & BIAS-VARIANCE TRADEOFF

Underfitting

High Bias / Low Variance

- Model is too simple
- Cannot capture patterns in data
- Poor performance on BOTH train and test
- Cause: too few features, too simple model
- Fix: More features, more complex model

Just Right ✓

Balanced Bias & Variance

- Model generalizes well
- Performs well on unseen data
- Good train AND test accuracy
- Goldilocks zone of complexity
- Achieved via cross-validation & tuning

Overfitting

Low Bias / High Variance

- Model is too complex
- Memorizes training data noise
- High train accuracy, low test accuracy
- Cause: too many features, deep model
- Fix: Regularization, dropout, more data

UNSUPERVISED LEARNING

FINDING PATTERNS WITHOUT LABELS

The algorithm explores unlabeled data to discover hidden structure, groupings, and patterns on its own — no predefined output labels are provided to guide training.

Clustering

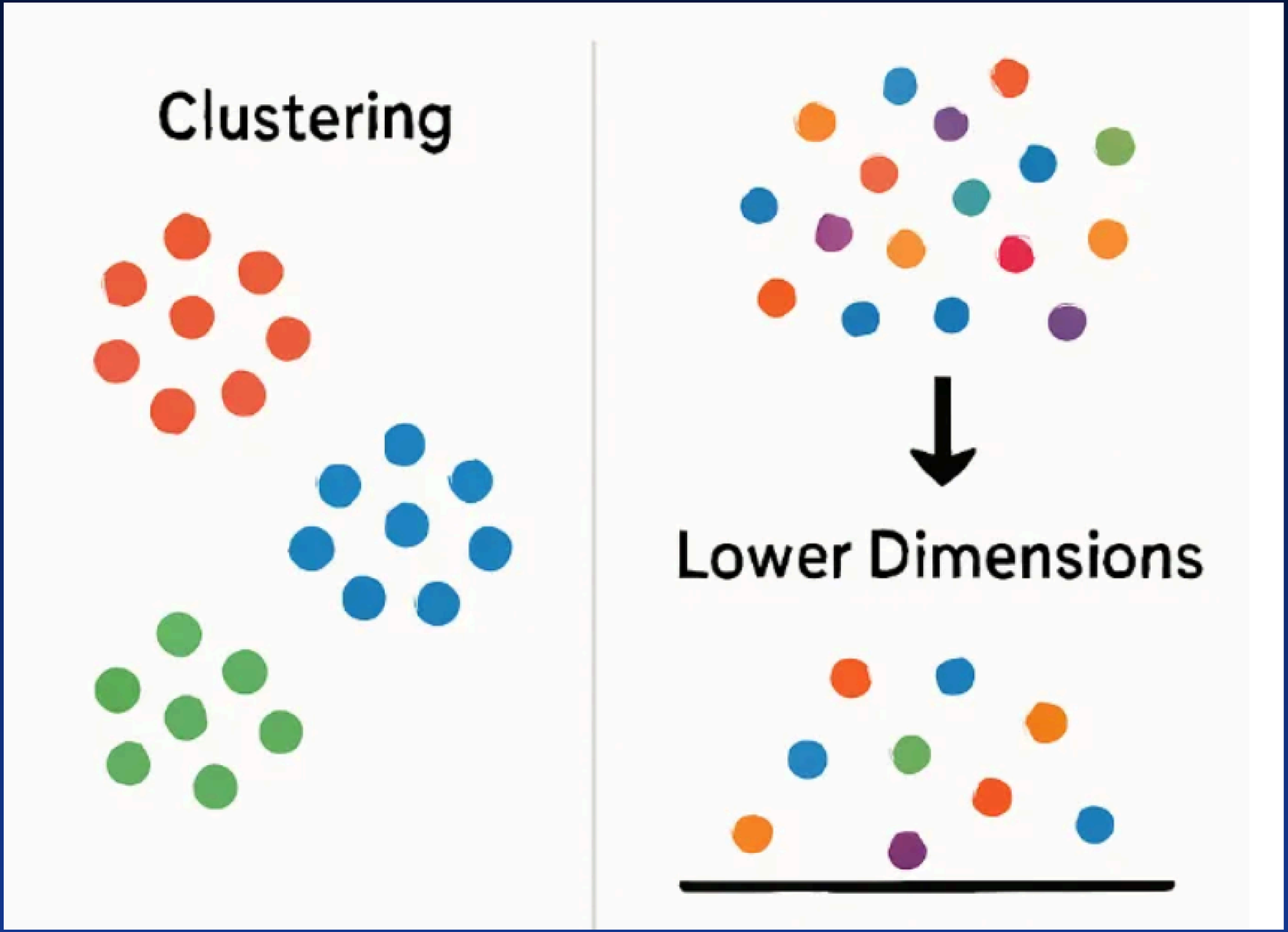
- Groups similar data points together
- No predefined class labels needed
- Discovers natural groupings in data
- Algorithms: K-Means, DBSCAN, Hierarchical, GMM, BIRCH
- Metrics: Silhouette Score, Inertia, Davies-Bouldin
- Use case: Customer segments, gene clustering, document topics

Dimensionality Reduction

- Reduces number of features in dataset
- Removes redundant or noisy information
- Makes data easier to visualize & process
- Algorithms: PCA, t-SNE, UMAP, Autoencoders, LDA
- Metrics: Explained variance, reconstruction error
- Use case: Image compression, feature extraction, NLP embeddings

UNSUPERVISED LEARNING

FINDING PATTERNS WITHOUT LABELS



UNSUPERVISED LEARNING

KEY UNSUPERVISED LEARNING ALGORITHMS

K-Means Clustering

Clustering

Partitions data into K clusters by minimizing distance to centroids. Iterates until convergence.

e.g. Customer segmentation by purchase behavior

DBSCAN

Clustering

Density-based: finds clusters of arbitrary shape; automatically marks outliers as noise.

e.g. Fraud detection, geographic hotspot analysis

Hierarchical Clustering

Clustering

Builds a dendrogram (cluster tree). No need to specify K. Can be agglomerative or divisive.

e.g. Gene expression analysis, social network grouping

PCA

Dim. Reduction

Principal Component Analysis: projects data to fewer dimensions while retaining max variance.

e.g. Face recognition preprocessing, image compression

t-SNE

Visualization

Visualizes high-dimensional data in 2D or 3D by preserving local cluster structure. Non-linear.

e.g. Visualizing word embeddings, single-cell genomics

Autoencoders

Both

Neural network encoder-decoder: compresses data to a latent space, then reconstructs it.

e.g. Image denoising, anomaly detection in manufacturing

Gaussian Mixture Model

Clustering

Soft clustering with probability assignments. Assumes data comes from a mix of Gaussians.

e.g. Speaker identification, image segmentation

UMAP

Dim. Reduction

Faster than t-SNE, preserves global structure better. Used for large datasets visualization.

e.g. Visualizing large text corpora, drug discovery

Apriori / Assoc. Rules

Association

Finds frequent itemsets and association rules in transaction data. Measures support & confidence.

e.g. Market basket analysis: "people who buy X also buy Y"

K-MEANS CLUSTERING: STEP-BY-STEP

1

Choose K

Choose how many groups you want.

Example:If K = 3, the computer will try to make 3 groups.

2

Initialize Centroids

A centroid is like the center of a group.The computer randomly places K center points on the map of your data.Think of it like putting 3 flags on the floor where the groups might form.

3

Assign Points

Each data point looks for the nearest centroid and joins that group.

Like every marble walking to the closest flag

4

Update Centroids

Now the computer moves each centroid to the middle of all the points in its group.

So the flag moves to the true center of its marbles.

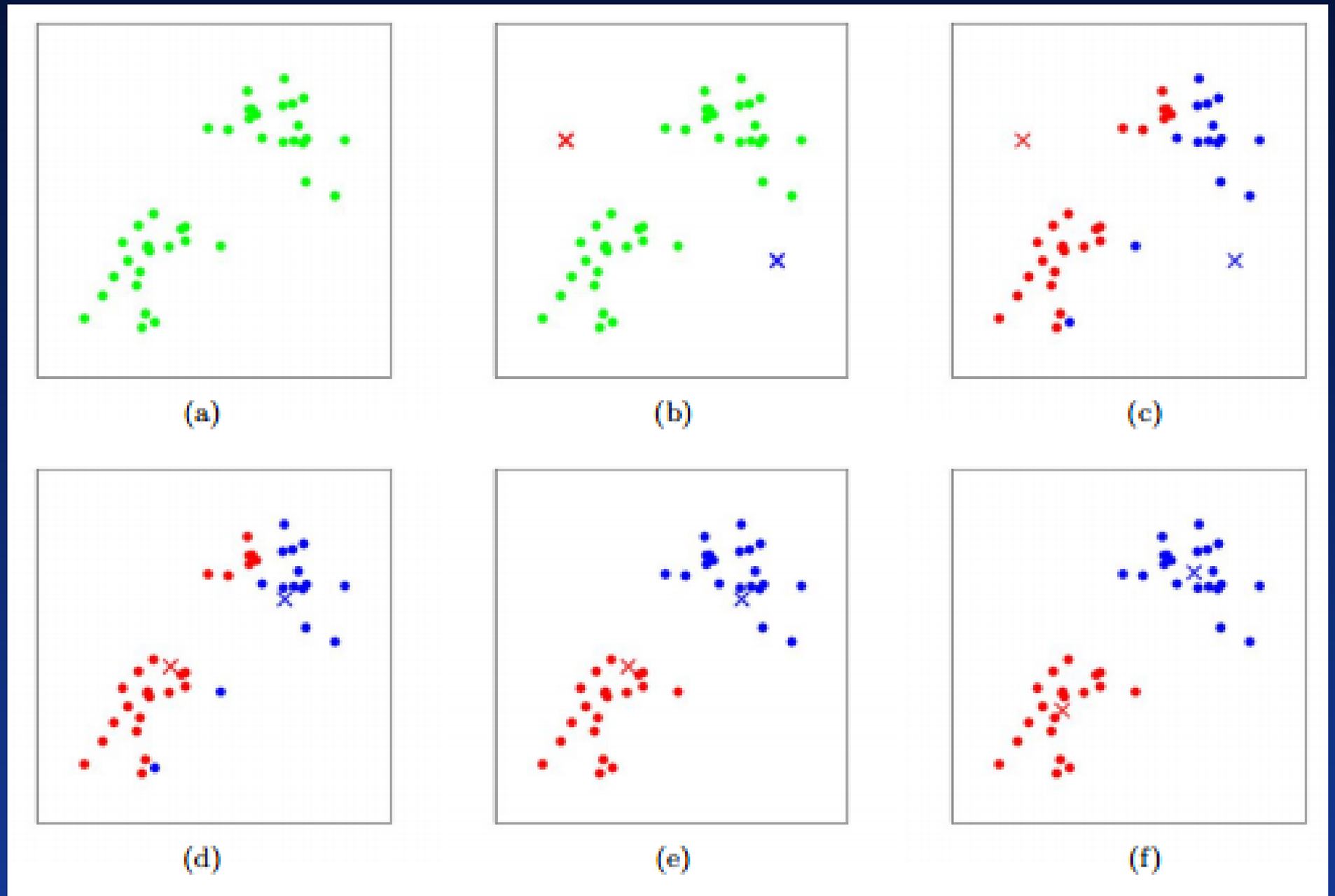
5

Repeat Until Convergence

The computer repeats:
- assign points
- move centroids

until nothing changes anymore (the groups stop moving).This is called convergence.

K-MEANS CLUSTERING: STEP-BY-STEP



SUPERVISED VS. UNSUPERVISED: FULL COMPARISON

Aspect	Supervised Learning	Unsupervised Learning
Labels Required	Yes — labeled data mandatory	No — raw unlabeled data used
Goal	Predict output for new input	Discover hidden patterns/structure
Output Type	Predictions / Classifications	Clusters / Encodings / Rules
Feedback Signal	Ground truth loss (e.g. MSE, cross-entropy)	Internal metric (e.g. inertia, silhouette)
Evaluation	Accuracy, F1, RMSE — clear & measurable	Subjective; harder to quantify quality
Data Cost	High — labeling requires human effort & time	Low — no labeling needed
Training Time	Moderate — depends on model and data size	Can be longer — finding structure is harder
Interpretability	Easier — predictions can be explained	Harder — clusters may need manual labeling
Common Algorithms	SVM, Decision Tree, Neural Net, Random Forest	K-Means, PCA, DBSCAN, Autoencoders
Real-World Example	Spam filter, credit scoring, disease diagnosis	Customer segments, topic modeling, anomaly detection

SUPERVISED VS. UNSUPERVISED: SIMPLE COMPARISON

Imagine teaching a kid animals with flashcards:

- Picture of a dog → label: “dog”
- Picture of a cat → label: “cat”
- Picture of a bird → label: “bird”

Labeled data means every example already has the correct answer attached.

You give a kid a big box of animal pictures but no names.
The kid might start grouping them like:

- furry animals together
- animals with wings together
- animals with fins together

Unlabeled data means there are no answers provided. The model tries to discover patterns by itself.

HOW DO YOU CHOOSE THE RIGHT APPROACH?

Do you have labeled data?

YES

Supervised Learning

Is the output a number?
→ Regression (Linear, Ridge, SVR)

Is the output a category?
→ Classification (Logistic, SVM, Tree)

NO

Unsupervised Learning

Want to find groups?
→ Clustering (K-Means, DBSCAN)

Want to simplify features?
→ Dim. Reduction (PCA, UMAP, t-SNE)

REAL-WORLD APPLICATIONS OF ML

Healthcare & Medicine

Supervised

Cancer detection from MRI scans (classification); drug efficacy prediction (regression)

Unsupervised

Patient clustering by symptoms; gene expression profiling; anomaly detection in vitals

Finance & Banking

Supervised

Credit scoring, loan approval, stock price prediction, insurance risk modeling

Unsupervised

Fraud transaction anomaly detection, customer segments, portfolio risk grouping

E-Commerce & Retail

Supervised

Product classification, demand forecasting, review sentiment analysis

Unsupervised

Customer segmentation for marketing, recommendation systems, market basket analysis

Natural Language Processing

Supervised

Spam filtering, sentiment analysis, named entity recognition, machine translation

Unsupervised

Topic modeling (LDA), word embeddings (Word2Vec), document clustering

Computer Vision

Supervised

Face recognition, object detection (YOLO), image captioning, OCR

Unsupervised

Image segmentation, feature extraction, generative models (GANs)

Autonomous Systems

Supervised

Traffic sign classification, obstacle prediction, path planning decisions

Unsupervised

Environment mapping, anomaly detection in sensor data, behavior clustering

THE ML PIPELINE & POPULAR TOOLS

Data Collection → Data Preprocessing → Feature Engineering → Model Training → Evaluation & Tuning → Deployment & Serving

Popular Libraries & Frameworks:

NumPy
Data

Numerical computation, arrays, linear algebra

Pandas
Data

Data manipulation, DataFrames, CSV/Excel handling

Matplotlib / Seaborn
Viz

Data visualization, plots, heatmaps, distributions

Scikit-learn
ML

ML algorithms: classification, regression, clustering, preprocessing, metrics

TensorFlow / Keras
DL

Deep learning framework by Google; neural networks, CNNs, RNNs

PyTorch
DL

Deep learning by Meta; dynamic computation graphs, research-friendly

XGBoost / LightGBM
ML

Gradient boosting; tabular data champion in competitions

Jupyter Notebook
Env

Interactive coding environment for data science workflows

Challenges, Limitations & Ethical Concerns

⚠️ Data Quality & Quantity

ML is only as good as its data. Biased, incomplete, or low-quality data leads to poor and unfair models. "Garbage in, garbage out."

🎭 Algorithmic Bias

Models trained on biased historical data will reproduce and amplify those biases. Example: facial recognition with lower accuracy for darker skin tones.

🔒 Privacy & Data Security

Training on sensitive data (health, finance) raises serious privacy concerns. Techniques: Differential Privacy, Federated Learning.

⬜ Explainability (Black Box)

Deep neural networks are hard to interpret. In healthcare and law, we need to explain WHY a model makes a decision. XAI (Explainable AI) addresses this.

🌍 Environmental Impact

Training large language models consumes massive energy. GPT-3 training emitted ~552 tons of CO_2. Efficient ML (TinyML, pruning) is an active research area.

📉 Overfitting & Generalization

A model that memorizes training data will fail in production. Robust generalization to new, out-of-distribution data remains an open challenge.

RECAP & KEY TAKEAWAYS

Machine Learning has evolved since 1950. From Turing's test to deep learning, decades of research built today's AI.

ML enables computers to learn patterns from data without being explicitly programmed for every case.

Supervised Learning uses labeled data to predict outcomes — split into Classification (categories) and Regression (numbers).

Unsupervised Learning discovers hidden structure in unlabeled data through Clustering and Dimensionality Reduction.

Choose based on your data: labeled data → Supervised; no labels → Unsupervised. Output type determines classification vs. regression.

ML powers real applications — healthcare, finance, NLP, autonomous vehicles, and much more. Ethical use is critical.

QUESTIONS?

ACTIVITY

1-5 Differentiate Supervised & Unsupervised Learning

Determine whether Supervised or unsupervised learning

6. A computer studies pictures labeled “apple,” “banana,” and “orange” so it can identify fruits in new pictures.
7. A program looks at thousands of songs and groups similar songs together based on sound patterns.
8. A model learns from past exam results labeled “pass” or “fail” to predict if a student will pass.
9. A computer analyzes shopping behavior and groups customers with similar buying habits.
10. A model studies images labeled “cat” and “dog” so it can tell them apart.